

DL4NLP – Lab 1 2026-2027

Task 1 (0.35p). Similarity depends on representation

Consider:

S1: The doctor examined the child.

S2: The physician examined the kid.

S3: The child examined the doctor.

- a) Build a set representation and a bag-of-words vector representation for the three sentences.
- b) Compute Jaccard similarity and cosine similarity for S1-S2 and S1-S3. Show the representations and the computations.
- c) Compute a token-level edit distance for the same pairs.
- d) Using WordNet, inspect the relations/similarity between doctor-physician and child-kid. Specify the synsets used.
- e) Which pair is semantically closer? Which representation(s) capture this correctly? Explain why S1 and S3 may appear deceptively similar in a bag-of-words representation.

Task 2 (0.65p). WordNet mini-challenge

(0.25p) Exploit WordNet's hierarchy using relations such as hyponym/hypernym, antonyms, meronyms, synonyms and definitions to collect words that are related to a user given word.

(0.25p) Build a Word Association Game using WordNet Similarity (any existing similarity method from NLTK library), along with the relations identified above. The player is prompted with a word and must come up with related words based on semantic similarity and relations using WordNet. Offer rewards points based on how semantically similar the player's word is to the original word. Provide feedback on the "closeness" of the player's word to the original.

(0.15p) Provide it in a user-friendly manner.